

Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!**

Part I. Do 2 of the following 3 problems. Set up integrals which solve the problems, but do NOT expend time computing the numerical values of these integrals

1. (12.5 pts) Figure 1 gives the vertical cross-section of tank which contains sea water (weight density = 64.0 lbs/ft^3), with depth 3 feet. Find the fluid force against the end of the tank.

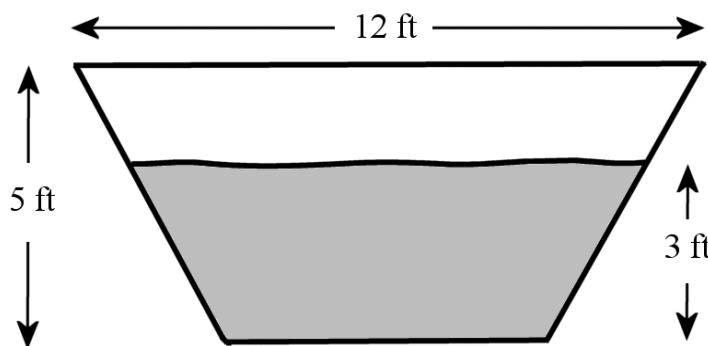


Figure 1

2. (12.5 pts) Figure 2 shows a trough which contains sea water (weight density = 64.0 lbs/ft^3), with depth 8 feet. How much work is done in pumping the sea water to the top of the trough?

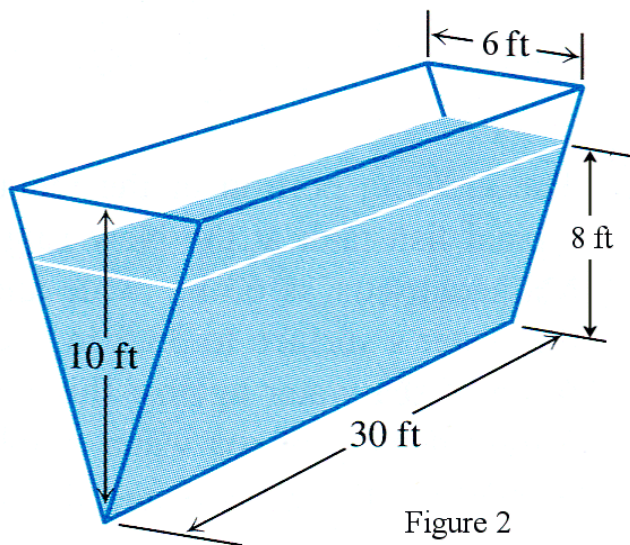


Figure 2

3. (12.5 pts) Find the coordinates of the centroid of the bounded region in the first quadrant bounded between the curves $y = x^2 + x$, $y = 0$, $x = 2$.

Part II. Do 8 of the following 9 problems

For each of the following problems find the value of the integral

4. (10 pts) $\int \frac{e^x}{3-4e^x} dx$

5. (10 pts) $\int 6xe^{2x} dx$

6. (10 pts) $\int \sin^3 2x \cos^4 2x dx$

7. (10 pts) $\int \frac{\sqrt{4+x^2}}{x^4} dx$

8. (10 pts) $\int \frac{x^2 + 2x - 9}{(x+1)^2(x-4)} dx$

9. (10 pts) $\int \frac{2x-12}{(x+1)(x^2+4x+5)} dx$

10. (10 pts) $\int \ln(2x+3) dx$

11. (10 pts) $\int_0^{\pi/3} \cos^2(6x) dx$

12. (10 pts) $\int_3^6 \frac{3x}{\sqrt{x-2}} dx$

Integrals

You are expected to have the following facts memorized.

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad (n \neq -1)$$

$$2. \int \frac{1}{x} dx = \ln |x| + C$$

$$3. \int e^x dx = e^x + C$$

$$4. \int a^x dx = a^x \frac{1}{\ln a} + C$$

$$5. \int \sin x dx = -\cos x + C,$$

$$\int \cos x dx = \sin x + C$$

$$6. \int \tan x dx = \ln |\sec x| + C,$$

$$\int \cot x dx = -\ln |\csc x| + C$$

$$7. \int \sec x dx = \ln |\sec x + \tan x| + C,$$

$$\int \csc x dx = -\ln |\csc x + \cot x| + C$$

$$8. \int \sec^2 x dx = \tan x + C,$$

$$\int \csc^2 x dx = -\cot x + C$$

$$9. \int \sec x \tan x dx = \sec x + C,$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$10. \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$$

$$11. \int \frac{1}{1+x^2} dx = \tan^{-1} x + C$$

$$12. \int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$13. \int \frac{1}{|x|\sqrt{x^2-1}} dx = \sec^{-1} x + C$$

$$14. \int \sinh x dx = \cosh x + C,$$

$$\int \cosh x dx = \sinh x + C$$